

Marginally Useful: Conformal Prediction Adds No Forecastability

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Abstract

The conformal step adds no forecastability. A conformal predictive system pools one law for its score across all inputs, and its log-score regret to the oracle is exactly the mutual information between score and input, plus a marginal mismatch that calibration removes. What the system knows about the future is therefore fixed before calibration runs, and it can be less than nothing. Pooling is a decision to stop transforming, and the same identity prices every stopping point. The conformal literature almost always stops at the pooled rank map, and on 572 economic time series that stop leaves a measurable gap which one scale transform removes.

1 The claim

Conformal prediction is sold to forecasters as the rigorous way to add uncertainty, the thing that “converts any statistical, machine or deep learning model into a probabilistic prediction model” (Manokhin, 2023); one forecasting library awards it alone the check mark for “Calibration Guarantee” (Nixtla, 2026). The guarantee is real: under exchangeability split conformal prediction returns a set with $\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha$, finite-sample and distribution-free (Vovk et al., 2005; Lei et al., 2018). But it is a statement about coverage, averaged over inputs, and a forecaster’s question is different: how much of what can be predicted does the method predict?

Under the log score that question has an exact form. The most any forecast can gain over the unconditional law of Y is the mutual information $I(Y; X)$ (DelSole, 2004), which Catt (2026) makes the definition of *forecastability*. We show that a conformal predictive system captures at most $I(Y; X) - I(T; X)$ of it, where T is the score it pools, and that the conformal step contributes none of it (Section 3). Pooling is a choice of where to stop transforming the data, the same identity prices every stopping point, and a coded survey of the literature shows the field stops at one point almost without exception (Section 4). On 572 economic series that stop costs a measurable amount, and one scale transform recovers it (Section 5).

2 Setup

Let (X, Y) have law P , with conditional density $p(y|x)$ and marginal density p_Y ; in time series X is the past and Y the value h steps ahead. The forecastability of Y given X is

$$F = \mathbb{E} \log p(Y|X) - \mathbb{E} \log p_Y(Y) = I(Y; X), \quad (1)$$

and the *exploitability* of a forecaster q is $\mathbb{E} \log q(Y | X) - \mathbb{E} \log p_Y(Y)$, which is at most F and can be negative (Catt, 2026).

A score transform is a family of strictly increasing C^1 bijections $\phi_x : \mathbb{R} \rightarrow \mathcal{Z}$ onto a common open interval, jointly measurable in (x, y) with their derivatives. Write $T = \phi_X(Y)$, with marginal law P_T and density p_T . The signed residual $\phi_x(y) = y - \hat{\mu}(x)$ gives the conformal predictive system (CPS) of Vovk et al. (2019); the normalized residual $(y - \hat{\mu}(x))/\hat{\sigma}(x)$ gives normalized conformal prediction (Lei et al., 2018); the estimated conditional rank $\Phi^{-1}(\hat{F}(y | x))$ gives distributional conformal prediction (Chernozhukov et al., 2021). The model is fitted on separate data and held fixed. A *pooled forecaster* chooses one density h on $\mathcal{Z}$ for every input and reports

$$q_h(y | x) = h(\phi_x(y)) \partial_y \phi_x(y). \quad (2)$$

A CPS is a pooled forecaster whose h is read off the calibration scores. Calibration sees the scores and never the inputs, so it chooses h and nothing else.

3 The gap

Theorem 1 (Information gap of a pooled forecaster). *Assume T has finite differential entropy, $I(T; X) < \infty$, and $\mathbb{E} |\log h(T)| < \infty$. Then for every density h on $\mathcal{Z}$,*

$$\mathbb{E} \log p(Y | X) - \mathbb{E} \log q_h(Y | X) = I(T; X) + \text{KL}(P_T \| H). \quad (3)$$

The regret is minimized at $h = p_T$, where it equals $I(T; X)$. No choice of h , and no recalibration that ignores X , brings it below $I(T; X)$.

Proof. With $t = \phi_x(y)$, $p(y | x) = p_T(\phi_x(y) | x) \partial_y \phi_x(y)$, so the Jacobians cancel against (2) and

$$\mathbb{E} \log \frac{p(Y | X)}{q_h(Y | X)} = \mathbb{E} \log \frac{p_T(T | X)}{p_T(T)} + \mathbb{E} \log \frac{p_T(T)}{h(T)} = I(T; X) + \text{KL}(P_T \| H),$$

each term finite under the assumptions. The second term vanishes only at $h = p_T$ (Gibbs). An X -blind recalibration applies a fixed increasing map ψ to the score, and $I(\psi(T); X) = I(T; X)$. $\square$

The identity is the chain rule for relative entropy along a product reference, familiar from universal coding as the split of redundancy into dependence and marginal mismatch (Clarke and Barron, 1990; Grünwald, 2007). What is new is the reading: it holds exactly, at every sample size, for the whole class every conformal predictive system belongs to. Reading the symmetric intervals of absolute-residual conformal as a distribution adds $\text{KL}(P_R \| H_{\text{sym}}) \leq \log 2$ for discarding the sign.

Lemma 1 (Which law the conformal system pools). *Let the calibration pairs be i.i.d. copies of (X, Y) , or stationary ergodic with that marginal law, and let $\hat{G}_n$ be the empirical distribution function of $T_i = \phi_{X_i}(Y_i)$. The signed CPS reports $\Pi_x(y) = \hat{G}_n(\phi_x(y))$ up to the $1/(n+1)$ fence-post convention, and $\sup_t |\hat{G}_n(t) - G_T(t)| \rightarrow 0$ almost surely (Glivenko–Cantelli, or the ergodic theorem on a countable dense set of t). If a smoothed version with density $\hat{h}_n$ has $\text{KL}(P_T \| \hat{H}_n) \rightarrow 0$, its regret tends to $I(T; X)$.*

Exchangeability alone does not give this: an infinite exchangeable sequence is a mixture of i.i.d. sequences, and $\hat{G}_n$ converges to a random limit. Coverage needs only exchangeability; the identification needs more, and the log score applies to a smoothed CPS because a finite empirical CPS has no Lebesgue density. Theorem 1 needs none of this, since it holds for every fixed h .

Corollary 1 (What a conformal system captures). *If also $I(Y; X) < \infty$, the pooled forecaster has exploitability*

$$\mathbb{E} \log q_h(Y | X) - \mathbb{E} \log p_Y(Y) = I(Y; X) - I(T; X) - \text{KL}(P_T \| H), \quad (4)$$

at most $I(Y; X) - I(T; X)$, a quantity fixed by the score before calibration. It is negative whenever $I(T; X) > I(Y; X)$.

Proof. Subtract (3) from (1). □

Remark 1 (Why the calibration gain is not forecastability). Calibration moves h toward p_T and removes $\text{KL}(P_T \| H)$. That term depends on the marginal law of T alone: a forecaster whose inputs carry no information gains it just as much. What depends on X is $I(Y; X) - I(T; X)$, and ϕ sets it. This is the sense in which the conformal step adds no forecastability. The negative case is ordinary: if $Y \perp X$ but $\hat{\mu}$ varies with x , the residual depends on X through the model alone, and a CPS with perfect coverage scores below the unconditional law by $I(R; X)$.

Remark 2 (Cells and conditional methods). A Mondrian CPS (Boström et al., 2021; Toccaceli, 2026) pools per cell $g(X)$ and has floor $I(T; X | g(X))$, so per-cell calibration captures $I(T; g(X))$ and no more. Distributional conformal prediction (Chernozhukov et al., 2021) has floor $I(T; X)$ for its estimated rank, zero when $\hat{F}$ is the true conditional law: it answers the problem through $\hat{F}$. Density-based, histogram, localized and quantile constructions (Izbicki et al., 2020, 2022; Sesia and Romano, 2021; Guan, 2023; Romano et al., 2019) likewise put X into the score or the reference class, and each lowers the floor by what that conditioning retains, never by the calibration step. Coverage and log score also grade different objects: two forecasters with the same $\hat{\mu}$ and different claimed densities have identical residual conformal sets while their log scores differ without bound, because the set never consults the density; Theorem 1 grades the density the CPS itself reports.

4 Pooling is a stopping decision

A forecaster can be written as a chain of transformations of the data: remove a location, divide by a scale, whiten serial dependence, map to ranks, and so on, each a state-dependent bijection. At some point the chain stops and one law is pooled for what is left. Split conformal prediction stops after the location, at the rank map of the residuals. Theorem 1 prices any stop: stopping at coordinates ϕ costs exactly $I(\phi_X(Y); X)$, and stopping at the true conditional rank costs nothing.

Two cautions keep this honest. The price need not fall as the chain grows, since an x -dependent map can raise $I(T; X)$ as well as lower it; the theorem prices each stop and does not say deeper is better. And the theorem is a population statement. With finite data an early stop regularizes, as early stopping does in estimation, so a stop can win series by series while losing on average.

Where the literature stops. We classified every paper on arXiv with “conformal prediction” in its title, 686 papers to August 2026, by what it does to the pooled rank map (protocol in Appendix A). A method can *stop* there and quote the pooled law; *replace* the pooled law with a conditional estimate; *tune* its level online; *consume* the conformal output in a downstream decision; or *compose*, keeping the map and fitting a model on its output.

move	papers	examples
stop	599	split conformal, CPS, CQR, Mondrian and normalized scores
replace	16	SPCI, localized conformal, HopCPT, EnbPI
tune	28	adaptive conformal inference, conformal PID
consume	38	conformal risk control, selection
compose	0	none found
no method	5	surveys, applications

The pattern is not an oversight of forecasting as a whole: mapping to empirical ranks and then continuing to model is standard in meta-Gaussian forecasting and copula time series (Bogner et al., 2012; Chen and Fan, 2006). It is a habit of the conformal literature, which knows the pooled law is the weak point and always swaps it out rather than keeping it and continuing.

5 Evidence

Exact checks. Both sides of (3) and (4) are computed independently by grid integration (`check_gap.py`) and agree to within 10^{-8} nats. With X uniform on $[0, 1]$ and $Y|x$ skew-normal with mean $2 \sin 3x$ and scale $0.5 + 1.5x$, $I(Y; X) = 0.340$. A correct location model leaves $I(R; X) = 0.118$ and so captures 65% of it; shrinking the location to $1.2 \sin 3x$ captures 56%. When $Y \perp X$ and $\hat{\mu}(x) = 1.5 \cos 4x$, the CPS captures -0.653 nats. A misspecified logistic $\hat{F}$ used as a distributional-conformal score gives $I(T; X) = 0.070$ against 0.123 for raw residuals under the same location: the score, not the calibration, halves the gap.

Coverage is blind to where the gap falls. With X uniform, $Y = \sigma(X)\varepsilon$, $\sigma(x) = 0.5 + 2x$, $\varepsilon \sim N(0, 1)$, raw and normalized split conformal share calibration sets of 500 and the rank index at $\alpha = 0.1$ (400 replications of 20,000 test points; standard errors at most 0.001):

	marginal	σ -quartile 1	2	3	4	width	interval score
raw	0.901	0.998	0.964	0.873	0.767	5.37	7.26
normalized	0.901	0.900	0.901	0.901	0.900	4.96	6.21

The certificate is identical in both rows. The difference is entirely the coordinates in which the pooling happens.

572 economic series. We use the continuous FRED change series (572 of 701 enumerated, the rest too short)¹ and a strong online forecaster, *laplace* (Cotton, 2026b), whose predictive mean is held fixed with the past as X . The oracle is unavailable, but the gap is observable from below: if M is the pooled law estimated from the past and Q any past-measurable challenger, then $\mathbb{E} \log(q/m) = \mathbb{E} \text{KL}(P \| M) - \mathbb{E} \text{KL}(P \| Q) \leq \mathbb{E} \text{KL}(P \| M)$, so a challenger’s held-out gain over the pooled law is, in expectation, a lower bound on what the pooled system pays the oracle. Pooling at four successive stops of the transform chain, with *laplace* itself run on the stream as challenger (nats per observation, series-bootstrap 90% intervals):

¹This table, the simulation above and the continuation result below first appeared in a working paper (Cotton, 2026a), which this article supersedes.

stop at	challenger gain	90% interval	positive on
raw residual (split conformal)	+0.117	[0.105, 0.129]	84% of series
standardized by an EWMA scale	−0.020	[−0.060, 0.015]	42%
serially whitened	−0.016	[−0.058, 0.017]	40%
PIT-normalized	+0.003	[−0.006, 0.012]	26%

Stopping where split conformal stops leaves at least 0.117 nats per observation. One scale transform leaves nothing this challenger can find. The calibration step is the same in every row.

Continuing past the rank map. The composing move that the survey finds unused does work. On the same series, keeping the rank map and fitting an AR(1) on its output beats stopping at it by 0.038 nats per observation on average (interval 0.027 to 0.052), and lowers the mean per-series CRPS ratio by 3.2%. The gain is skewed: the median series is a coin flip, and on i.i.d. null series the comparison reverses by 0.011 to 0.051 nats, the estimation cost of refitting when there is nothing to find. Stopping is defensible series by series and beaten on the corpus, as Section 4 anticipates.

6 What this does not say

The coverage guarantee stands, and where coverage is the deliverable (selective prediction, shortlisting, conformal p -values) conformal prediction is the right tool (Angelopoulos et al., 2021; Bates et al., 2021). Within the pooled class the CPS is log-score optimal; the cost is the class. We use *coverage* for the property of a set and *calibration* only for the distributional property (Gneiting et al., 2007). The gap is driven by how the law of T varies with x , not by whether x is typical under p_X . The information identities belong to the log score; for CRPS or the interval score the analogous floor is not an information quantity and we claim no identity for it. Finally, the order matters: a Bayesian predictive or other conditional model can supply the distribution and conformal calibration can certify it, and then the certificate costs no forecastability if the model has made T nearly independent of X . Conformalize last, and put the conditioning in the score.

Related work. The CPS (Vovk et al., 2019), the calibration–sharpness paradigm (Gneiting et al., 2007), the impossibility of distribution-free conditional coverage (Lei and Wasserman, 2014; Foygel Barber et al., 2021), information-theoretic readings of set size (Correia et al., 2024), the observations that validity invites weak predictors (Recht, 2024) and that length can be gamed at fixed coverage (Min et al., 2026), and forecastability as mutual information (DelSole, 2004; Catt, 2026) are all prior. To our knowledge the exact regret of the pooled class, its identification with the conformal predictive system, the resulting account of the forecastability conformal prediction captures, and the pricing of the stop the literature makes have not been stated before.

7 Conclusion

Coverage answers *is the truth in this region?* A forecaster asks how much of what can be predicted has been predicted. For conformal prediction the answer is $I(Y; X) - I(T; X)$ at best, set by the model and the score, with the calibration step contributing nothing that depends on the inputs. The price of the coverage guarantee, paid in the forecaster’s own currency, is $I(T; X)$, and it is the price of stopping where the field stops.

Data availability. The FRED series list, the survey frame with its labels, and all code are supplied with the submission. `check_gap.py` reproduces every exact check.

A Survey protocol

To be completed from papers/grammar/survey/ before submission. Frame: arXiv API query `ti:"conformal prediction"`, all dates to August 2026, 686 records. Screen: documented keyword rules assign a provisional label and flag every possible *compose* paper; a missed compose is the only error that damages the claim, so the flag is deliberately loose. Reading: all flagged papers read in full text (7, none composes), and a random validation sample of N of the remainder hand-coded to estimate the screen's error rate on the other labels. Category rules, with the sentence from each flagged paper that places it, are in the supplementary table.

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